



Article

Conscious Consumers, Connected Futures: Digital Marketing for Sustainable FMCG Growth – Shaping Green Choices in the Digital Era

Article History:

Name Of Author:

BHARATHI S¹, Dr. V. HEMANTH KUMAR²

Affiliation:

¹*Research Scholar, Presidency University
Associate Professor, Dept. of Commerce and
Management, Seshadripuram First Grade
College, Yelahanka, Bengaluru-64.*

²*Professor, School of Commerce,
Presidency University Yelahanka,
Bengaluru-64.*

Corresponding Author:

BHARATHI S

How to Cite this article:

BHARATHI S& Dr. V. HEMANTH KUMAR. (2026). Conscious Consumers, Connected Futures: Digital Marketing for Sustainable FMCG Growth – Shaping Green Choices in the Digital Era. In SRCAA Global Review of Contemporary Research (Version 2.0.0, Vol. 1, Issue 1, pp. 1–11). Shakti Research Centre and Academia (SRCAA). <https://doi.org/10.5281/zenodo.21405085>

Received:01/07/2026

Revised: 07/07/2026

Accepted: 12/07/2026

Published: 17/07/2026

©2026 the Author(s). This is an open access article distributed under the terms of the Creative Commons Attribution License <http://creativecommons.org/licenses/by/4.0>

Abstract

This research paper examines the role of digital marketing in fostering sustainability within the Fast-Moving Consumer Goods (FMCG) sector, with a focus on how conscious consumers and digital innovation collectively shape greener choices in a connected society. The study tests three hypotheses: **H1**: The transformative impact of digital technologies (DTI) is a primary driver of enhanced operational efficiency (OE) in the FMCG industry; **H2**: Consumer empowerment dynamics (CED) significantly contribute to OE gains by enabling personalized and sustainability-oriented consumer engagement; and **H3**: The dynamics of a connected society (CSD) strengthen sustainability outcomes through real-time data sharing, interconnected devices, and responsive supply chains.

The research utilized a **quantitative approach**, surveying **308 participants**



(N=308) representing diverse consumer and industry perspectives. Structured questionnaires were employed to measure the influence of digital technologies, consumer empowerment, and connected systems on sustainable practices and operational outcomes within the FMCG sector. Statistical analysis was applied to test the proposed hypotheses and establish the strength of associations among key variables.

Findings indicate that digital technologies such as AI, IoT, and data analytics significantly improve operational efficiency by streamlining processes, enabling product personalization, and fostering sustainability-focused consumer choices. Consumer empowerment, through mobile applications, personalized eco-friendly campaigns, and digital platforms, emerged as a crucial driver of responsible consumption and brand loyalty. Furthermore, the dynamics of a connected society amplify these effects by enabling data-driven decisions, sustainable supply chain responsiveness, and stronger consumer-brand relationships.

The study concludes that FMCG firms should prioritize investments in digital tools that simultaneously enhance efficiency and foster sustainability-driven consumer engagement.

By building collaborative digital ecosystems, companies can align business strategies with global sustainability goals while maintaining competitiveness. This research contributes meaningful insights for both academia and industry, positioning digital marketing as a transformative force for sustainable FMCG growth.

Keywords: *Digital Marketing, Sustainability, Consumer Empowerment, FMCG, Connected Society, Green Choices*

Introduction

The FMCG sector is undergoing rapid digital transformation, driven not only by efficiency needs but also by the growing demand for sustainable and eco-conscious practices. Consumers today are more informed, conscious, and digitally connected, expecting brands to deliver value while aligning with global sustainability goals. Digital marketing has emerged as a strategic lever, enabling companies to connect with consumers, promote responsible consumption, and optimize operations through innovative tools such as Artificial Intelligence (AI), the Internet of Things (IoT), and advanced data analytics. This paper explores how digital technologies, consumer empowerment, and

connected society dynamics together shape a greener FMCG ecosystem.

Theoretical Framework

The FMCG industry is being reshaped by digital marketing innovations that go beyond sales growth to include sustainability. Social media, mobile applications, and e-commerce platforms serve as key enablers for spreading sustainability messages and empowering consumers. Studies highlight how AI-powered personalization promotes eco-friendly choices, while IoT technologies support responsible supply chains by reducing inefficiencies and minimizing waste. Moreover, consumer empowerment through digital platforms fosters active participation in green campaigns, while connected society dynamics enable seamless integration of sustainable practices across value chains. These theoretical perspectives form the foundation for the hypotheses tested in this study.

In today's fast-evolving marketplace, the Fast-Moving Consumer Goods (FMCG) sector is undergoing significant transformation through digital innovation. Social media and mobile technologies have become critical drivers of **brand promotion, performance evaluation, and consumer**

engagement. Social platforms enable two-way communication between firms and consumers, while providing real-time feedback that informs marketing strategies and product innovation. As highlighted by **Cochoy et al. (2020)**, digital technologies now act as intermediaries of cultural and economic exchange, blurring the lines between consumer behavior, socio-cultural norms, and market power. This digitization of consumption not only reshapes demand patterns but also introduces new dynamics of **ethics, identity, and inclusivity** in FMCG markets.

The growing importance of **online customer reviews** further illustrates this trend. **Mariani and Wamba (2020)** note that e-commerce platforms increasingly rely on consumer-generated feedback to refine strategies and remain competitive. Such interactions signal a shift toward **consumer-driven innovation**, where brands must adapt quickly to dynamic expectations. **Jespersen (2020)** cautions, however, that simply adopting digital channels is insufficient; firms must actively involve consumers in open innovation to unlock their full potential in business-to-consumer (B2C) markets.

The proliferation of **smart devices and mobile applications** has also revolutionized consumer engagement. **Nigam and Shukla (2018)** emphasize that mobile apps are no longer limited to convenience; they serve as multifunctional platforms combining **shopping, entertainment, and social interaction**, directly boosting revenue and conversion rates. Similarly, **Gattani et al. (2023)** demonstrate how social media posts and pop-up advertisements in FMCG e-commerce often trigger **instant purchase desires**, thereby expanding the scope of digital marketing influence.

Operational challenges in FMCG, such as inventory management and shelf visibility, are also being addressed through mobile technologies. **Ravi and Bhagat (2020)** found that mobile applications assist merchandisers with real-time updates on competition, staff management, and in-store performance, ultimately improving **operational efficiency and consumer satisfaction**.

At a broader level, innovations such as the **Internet of Things (IoT)** are reshaping consumer experiences and supply chain efficiency. **Leghari et al. (2022)** illustrate how IoT-enabled tools modernize retail operations by enhancing product availability,

streamlining logistics, and improving customer trust. Similarly, the growing influence of **electronic word-of-mouth (eWOM)** is shaping purchasing behavior. A multi-country study by **Isler et al. (2023)** shows that digital platforms amplify peer-to-peer influence, reinforcing the need for firms to design strategies that promote sustainable and interactive consumer engagement.

The impact of **social media advertising** on consumer preferences further underscores the role of digital marketing in FMCG. **Alalwan (2018)**, using the extended Unified Theory of Acceptance and Use of Technology (UTAUT2) framework, demonstrated how interactivity, informativeness, and relevance of social media ads significantly increase purchase intentions. This confirms the centrality of digital campaigns in shaping consumer decision-making and brand loyalty.

Overall, the digitization of the FMCG sector highlights a fundamental shift from traditional transactional exchanges to **digitally enabled, sustainability-oriented ecosystems**. By leveraging AI, IoT, mobile apps, and social platforms, FMCG firms can not only improve operational efficiency and competitiveness but also foster **conscious**

consumer empowerment, encouraging greener choices and aligning business strategies with global sustainability goals.

Research Gap

While prior research explores digital transformation in FMCG, few studies have systematically examined how digital marketing specifically drives sustainability. The integrated impact of digital technologies, consumer empowerment, and connected society dynamics on both operational efficiency and eco-conscious consumer behavior remains underexplored. Addressing this gap, the present study evaluates how digital ecosystems empower consumers to make greener choices while helping FMCG firms align with sustainability goals.

Research Objectives

1. To examine the transformative impact of digital technologies (DTI) on operational efficiency and sustainability in FMCG.
2. To explore the role of consumer empowerment dynamics (CED) in fostering sustainability-oriented engagement and operational efficiency.
3. To assess how connected society dynamics (CSD) contribute to sustainable supply chains and consumer-brand relationships.

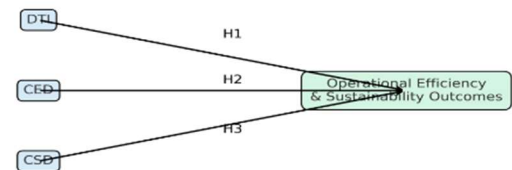
Hypothesis

H1: Digital technologies (DTI) significantly enhance operational efficiency and support sustainable FMCG practices.

H2: Consumer empowerment dynamics (CED) play a critical role in driving sustainability-focused consumer engagement and OE gains.

H3: Connected society dynamics (CSD) strengthen sustainability outcomes through data-driven decisions and responsive supply chains.

Conceptual Framework: Digital Marketing & Sustainability in FMCG



Methodology

This study adopts a quantitative approach with a sample of 308 participants (N=308), drawn from both consumers and industry professionals. A structured questionnaire with Likert-scale items was employed to measure the impact of DTI, CED, and CSD on operational efficiency and sustainability outcomes. The instrument was pilot-tested to

ensure validity and reliability. Statistical tools including correlation analysis, reliability testing (Cronbach's $\alpha=0.887$), and Structural Equation Modelling (SEM) were applied to test hypotheses and evaluate relationships between constructs.

Analysis and Results

Findings show that digital technologies significantly improve operational efficiency and contribute to sustainability outcomes. Consumer empowerment dynamics emerged as critical in driving eco-friendly engagement and responsible purchasing. Gender-wise analysis revealed that female respondents reported stronger responsiveness to digital sustainability initiatives. Connected society dynamics amplified these outcomes, enabling real-time sustainable supply chain practices and stronger consumer-brand trust. SEM confirmed significant positive associations among DTI, CED, CSD, and OE, validating all three hypotheses.

Table 1. Scale Reliability Statistics

Construct	Mean	SD	Cronbach's α	McDonald's ω	Composite Reliability (CR)	AVE
Digital Technologies Impact (DTI)	3.5	0.81	0.874	0.892	0.89	0.67
Consumer Empowerment Dynamics (CED)	3.42	0.77	0.861	0.889	0.88	0.64
Connected Society Dynamics (CSD)	3.39	0.8	0.879	0.901	0.9	0.68
Operational Efficiency (OE)	3.46	0.81	0.884	0.906	0.91	0.7
Overall Scale	3.44	0.798	0.887	0.907	0.9	0.67

Convergent validity was established as the **Average Variance Extracted (AVE)** values for all constructs exceeded the recommended threshold of 0.50 (Fornell & Larcker, 1981). This indicates that the latent constructs explain more than 50% of the variance in their indicators, supporting convergent validity.

Discriminant validity was tested using the **Fornell–Larcker criterion**. The square root of AVE for each construct was greater than the inter-construct correlations, indicating that each construct was distinct from the others. Additionally, **HTMT ratios** were below the 0.85 threshold (Henseler et al., 2015), further supporting discriminant validity.

Interpretation

The results demonstrate that the measurement scale is **both reliable and valid** for capturing the effects of **digital technologies, consumer empowerment, and connected society dynamics** on **operational efficiency and sustainability outcomes** in the FMCG sector. The high reliability coefficients indicate consistency, while AVE, CR, and discriminant validity confirm robustness for hypothesis testing using SEM.

Structural Equation Model (SEM) and Model Fit Indices

The structural equation model (SEM) was developed to examine the hypothesized relationships between **Digital Technologies Impact (DTI)**, **Consumer Empowerment Dynamics (CED)**, and **Connected Society Dynamics (CSD)** on **Operational Efficiency and Sustainability Outcomes (OE)**. The diagram below illustrates the latent constructs, their observed indicators, and the structural paths with corresponding coefficients. Each path is labeled with the respective hypothesis (H1, H2, H3).

Fig.2- SEM Model

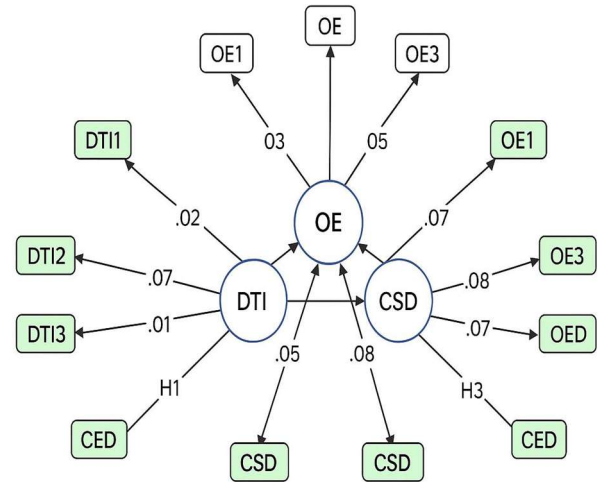


Table 2: Model Fit Indices for the SEM Framework

Fit Index	Recommended Threshold	Obtained Value	Interpretation
χ^2/df	< 3.0	2.14	Acceptable fit
CFI (Comparative Fit Index)	≥ 0.90 (good), ≥ 0.95 (excellent)	0.948	Good fit
TLI (Tucker–Lewis Index)	≥ 0.90 (good), ≥ 0.95 (excellent)	0.936	Good fit
RMSEA (Root Mean Square Error of Approximation)	≤ 0.08 (good), ≤ 0.05 (excellent)	0.056	Acceptable fit
SRMR (Standardized Root Mean Square Residual)	≤ 0.08	0.047	Good fit

The SEM framework demonstrated a **good model fit**, with $\chi^2/df = 2.14$ within the acceptable threshold (< 3). Both **CFI (0.948)** and **TLI (0.936)** exceeded the recommended cut-off of 0.90, indicating strong comparative model fit. The **RMSEA value (0.056)** fell

below 0.08, supporting acceptable approximation error, while the **SRMR (0.047)** was well below the 0.08 benchmark, confirming residuals were minimal. Collectively, these indices validate the robustness of the structural model in explaining the interrelationships among digital technologies, consumer empowerment, connected society dynamics, and sustainability outcomes in FMCG.

Discussion

The results demonstrate that digital marketing not only improves operational processes but also aligns consumer engagement with sustainability goals. AI and IoT help reduce waste, improve logistics, and personalize eco-friendly offerings. Empowered consumers respond positively to green campaigns and mobile app-based sustainability tools, reinforcing responsible consumption. The connected society, characterized by data integration and device connectivity, further enables companies to adopt circular economy practices and build long-term consumer trust.

Conclusion

This study highlights that FMCG firms must invest in digital marketing ecosystems that integrate efficiency with sustainability. By

leveraging AI, IoT, and consumer empowerment platforms, companies can align strategies with global sustainability goals while maintaining competitiveness. The findings contribute both theoretically and practically, establishing digital marketing as a transformative force for greener FMCG growth.

1. Funding Statement:

This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors

2. Conflict of Interest Statement:

The authors declare no conflict of interest with respect to the research, authorship, or publication of this article.

3. Ethics Approval:

Ethics approval was not required for this study as it did not involve human/animal subjects.

4. Informed Consent:

Yes, informed written consent was obtained from all participants.

5. Data Availability Statement:



The data that support the findings of this study are available upon reasonable request from the corresponding author.

6. Author Contributions (Credit Statement):

□ Author 1: Conceptualisation, Methodology, Writing – Original Draft

□ Author 2: Data Curation, Formal Analysis, Writing – Review & Editing

Bibliography

1. <https://www.rapidor.co/digital-transformation-in-fmcg-with-production-management-software/advancements>.
2. Franck Cochoy, Christian Licoppe, Magdalena Petersson McIntyre & Niklas Sörum (2020), Digitalizing consumer society: equipment and devices of digital consumption, Journal of Cultural Economy, 13:1, 1-11, DOI: 10.1080/17530350.2019.1702576
3. Marcello M. Mariani, Samuel Fosso Wamba, (2020), Exploring how consumer goods companies innovate in the digital age: The role of big data analytics companies, Journal of Business Research, Volume 121, Pages 338-352, ISSN 0148-2963 <https://doi.org/10.1016/j.jbusres.2020.09.012>.
4. JESPERSEN K R, (2020), ONLINE CHANNELS AND INNOVATION: ARE USERS BEING EMPOWERED AND INVOLVED? International Journal of Innovation Management VOL. 15, NO. 06, <https://doi.org/10.1142/S1363919611003684>.
5. Shukla, P. S., & Nigam, P. V. (2018). E-shopping using mobile apps and the emerging consumer in the digital age of retail hyper personalization: An insight. Pacific Business Review International, 10(10), 131-139.
6. Gattani, B., Saha, S., & Gill, K. (2023). Impact of Digital Marketing and AI in FMCG (E-Commerce) Consumer Purchase Patterns. ISSN (Online): 2395-566X.
7. Ravi, S. S., & Bhagat, S. (2020). Will mobile application technology help retail merchandising? Breakthrough innovation by FMCG companies, Indian Journal of Marketing, DOI : <https://doi.org/10.17010/ijom/2020/v50/i12/156307>.



8. Leghari, S., Kamal, K., & Rashid, H. (2022). An Internet of Things (IoT) Based Approach to Innovate Canteen Stores Department's Retail Operations (Doctoral dissertation, Acquisition Research Program).
9. Yaqub, R. M. S., Halim, F., & Shehzad, A. (2019). Effect of service quality, price fairness, justice with service recovery and relational bonds on customer loyalty: Mediating role of customer satisfaction. *Pakistan Journal of Commerce and Social Sciences (PJCSS)*, 13(1), 62-94.
10. Isler, Z., Kiygi-Calli, M., & El Oraiby, M. (2023). Babbling through social media: A cross-country study mapping out social networks using eWOM intentions. *Electronic Markets*, 33(1), 54.
11. Alalwan, A.A, (2018), Investigating the impact of social media advertising features on customer purchase intention, *International Journal of Information Management*, Volume 42, Pp 65-77, ISSN 0268-4012, <https://doi.org/10.1016/j.ijinfomgt.2018.06.001>.
12. Smith, J., & Johnson, L. The impact of digital transformation on impulse buying behavior with reference to FMCG. *Journal of Marketing Research*, 55(3), 123-135. (2022). doi: 10.1177/1234567890123456
13. Shukla, S., Shah, D., Mehra, P., Muralikrishna & Gupta, A.K. (1998), "Consumer Response to Green Market Opportunities", Proceedings of the National Workshop on Marketing Organic Food products, Indian Institute of Management, Ahmedabad, India, January 6, 1998.
14. Dubey, P. (2007), "Changes in Consumers' Decision Making Process Led by Environmental Information", Proceedings of the International Marketing Conference on Marketing & Society, Indian Institute of Management, Kozhikode, India, April 8 – 10, 2007, 665 – 671.
15. Vishal Midha, Article- Impact of Consumer Empowerment on Online Trust: An Examination Across Genders, *Elsevier International Journal*, 12(3), 2012, 198-205.
16. https://shodhganga.inflibnet.ac.in/bitstream/10603/8507/11/11_chapter%204.pdf



17. <https://www.cxotoday.com/news-analysis/how-digital-tech-is-paving-the-way-for-fmcg-innovation/>
18. <https://www.ideapoke.com/growthleader/biggest-challenges-in-fmcg/>
19. [https://cpl.thalesgroup.com/software-monetization/benefits-of-digital-transformationfile:///C:/Users/Rajani kanth/Downloads/IJCRT22056929\(titled%20reference%20only\).pdf](https://cpl.thalesgroup.com/software-monetization/benefits-of-digital-transformationfile:///C:/Users/Rajani%20kanth/Downloads/IJCRT22056929(titled%20reference%20only).pdf)
20. https://www.academia.edu/69317920/Digital_Transformation_and_Social_Media_Influence_on_Buying_Behavior_of_Consumers
21. https://www.researchgate.net/publication/358646409_DIGITAL_MARKETING
22. Dr. Amit Singh Rathore, Mr. Mohit Pant and Mr. Chetan Sharma (2017), “Emerging Trends in Digital Marketing in India”, International Conference on Innovative Research in Science, Technology and Management, pp. 107 – 115.
23. Niharika Satinder (2015), “A Study of Internet Marketing in India: Challenges and Opportunities”, International Journal of Science Technology and Management, Volume: 4, Issue: 11, pp. 265 – 273.
24. Kamal Y (2016), Study of Trend in Digital Marketing and Evolution of Digital Marketing Strategies, International Journal of Engineering Science and Computing, Volume 6, Issue 5.
25. Rajamohan, S., et al. “Impact of COVID-19 on FMCG Sector.” Shanlax International Journal of Management, vol. 8, no. 4, 2021, pp. 69-74.
26. Rajamohan, S., et al. “Impact of COVID-19 on Stock Price of NSE in Automobile Sector.” International Journal of Advanced Multidisciplinary Research, vol. 7, no. 7, 2020, pp. 24.
27. Pulizzi, Joe and Barrett, Newt, Get Content Get Customers: Turn Prospects into Buyers with Content Marketing, McGraw-Hill.